

DAY 3 · SESSION 6

FinOps for Research Data, Data Management as Code, and the Roadmap Capstone

Azure Storage, Data Protection, and Resilience at Scale - University of Minnesota

This afternoon, block by block

01 **Block 1 · 55 min**

Storage cost anatomy, optimization levers, and FinOps operating model

02 **Block 2 · 50 min**

Data management as code, governance guardrails, and the platform demonstration

03 **Break · 15 min**

Reset before the capstone

04 **Block 3 · 105 min**

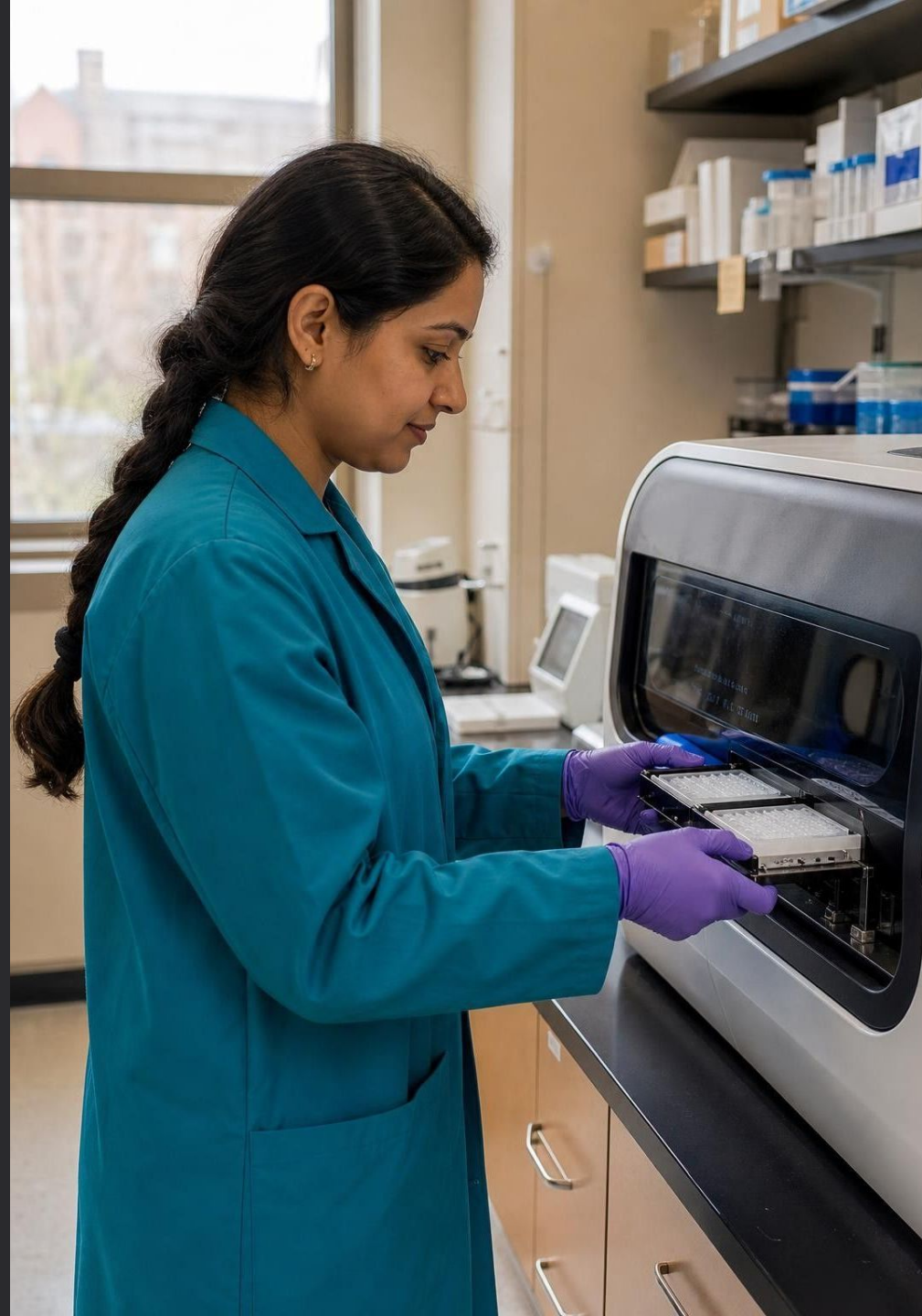
Capstone roadmap workshop and presentations

05 **Block 4 · 15 min**

Course wrap-up and survey

Low unit costs decide whether research data can be protected at all

- FinOps is a critical requirement: manage a multi-petabyte portfolio economically and justify the cost of protection
- Unit costs must stay low enough that irreplaceable but non-critical research data can be protected
- The team is automation driven and expects storage, protection, and policy as code



By the end of this session you can



Model capacity, transaction, egress, rehydration, and early deletion costs without surprises



Apply reserved capacity, tiering, and lifecycle automation without reducing protection



Implement showback and chargeback with tags, allocation, budgets, and anomaly alerts



Author Bicep modules and enforce standards with Azure Policy



Draft a two-year hybrid data platform roadmap

Every session had a cost line: today we add them up

01

Session 1

Tiers and lifecycle policies

02

Session 2

Archive economics and egress

03

Session 3

Backup retention and vault redundancy

04

Session 4

The cost of each resilience tier

05

Session 5

The business case for immutable database protection

BLOCK 1

Storage cost anatomy, optimization levers, and FinOps operating model



The anatomy of a storage bill



Capacity

GB stored per month, priced by tier and redundancy



Transactions

Write and read operations, higher in cooler tiers



Data retrieval

Per-GB charges reading from Cool, Cold, and Archive



Egress

Data leaving the region



Rehydration and early deletion

The surprise line items

Estimating capacity cost the Microsoft way

1

Base capacity

Multiply data size in GB by the tier's storage price

2

Access pattern

Add transaction, retrieval, and transfer estimates for the access pattern

⚠ Microsoft's estimation article uses sample prices only; use the official pricing page or the pricing calculator for real figures.

The most common surprises

Early deletion

Charged when data leaves Cool (30 days), Cold (90), or Archive (180) too soon

Rehydration priority

High-priority rehydration costs more than standard

Auto-tier to Hot

Can trigger early deletion fees

Small objects

Millions of them multiply transaction and tier-change costs

Hidden capacity

Versions, snapshots, and soft-deleted data are all billed

Smart Tier

Removes early deletion and retrieval charges inside Hot, Cool, and Cold and adjusts billing

Research data and enterprise data cost differently



Research

Huge capacity, low access, long retention; capacity and Archive pricing dominate



Enterprise

Smaller capacity, heavy transactions, tight RPO and RTO; redundancy, compute, and backup dominate

Optimize each with different levers, and report them separately.

Reserved capacity: commit and save

1 or 3 yr

Discount on block blob and Data Lake Storage capacity in standard accounts

100 TiB

Purchased in units of 100 TiB and 1 PiB per month

3 tiers

Available for Hot, Cool, and Archive

- Available for Hot, Cool, and Archive in standard accounts; not for Cold, smart tier, or premium accounts
- Bound to one region, one access tier, and one redundancy option, so plan the portfolio by those three dimensions before you buy
- Scope to one subscription, shared subscriptions, or a management group; it is a billing discount only and changes nothing about the resources
- Capacity only: transactions, egress, retrieval, and early deletion are not covered

More levers that keep protection intact

Tiering

Tier and lifecycle tuning based on measured access

Redundancy

Right-sizing per data class (Session 4 matrix)

Retention

Keep what policy requires, not more

Smart tier

Automatic Hot, Cool, and Cold movement for research data with unpredictable access

Files provisioned v2

Provision storage, IOPS, and throughput separately for departmental shares

Use **Azure Advisor** cost recommendations as a monthly input to the review cadence.

FinOps in one cycle: inform, optimize, operate



Inform

Allocate and report costs to research units



Optimize

Act on tiering, reservations, and rightsizing



Operate

Budgets, alerts, and a monthly review cadence

Aligned to the FinOps Foundation framework.

Microsoft Cost Management capabilities

Analyze

Report and analyze costs in the Azure portal or Power BI

Alert

Budget, anomaly, reservation utilization, and scheduled alerts

Allocate

Tag inheritance and cost allocation rules to split shared costs

Export

Exports to automate processes and feed external tools

Showback and chargeback for research units

Tagging standard: research unit, project, data class, retention class, owner

01

Inherit tags

Resource costs carry subscription and resource group tags

02

Allocate

Split shared platform costs with allocation rules

03

Budget alerts

Notify recipients when cost reaches the defined threshold

04

Anomaly alerts

Flag unexpected spikes automatically

BLOCK 2

Data management as code, governance guardrails, and the platform demonstration



Why data management as code



Consistent

Every account deployed identically: encryption, network rules, lifecycle, immutability, tags



Reviewable

Changes flow through pull requests



Faster

Safer provisioning for faculty and research requests



Self-documenting

The template is the documentation

Bicep modules for storage and protection



Storage account module: SKU, HNS, network rules, minimum TLS, shared key setting



Lifecycle management policy as a child resource



Container immutability and versioning settings



Backup vault with immutability, plus RBAC role assignments and tags



Parameterize per data class; reuse across research units

Deployment stacks: manage the platform as one unit

One unit

A deployment stack manages a group of Azure resources as a single unit

actionOnUnmanage

Resources removed from the template are detached or deleted

Deny settings

Block unwanted changes to managed resources

Subscription scope

Include the resource group, since deleting a resource group bypasses deny assignments

Pipeline pattern: from commit to compliant platform

1

Commit

Bicep changes to the repository

2

Validate

Lint and build, then a what-if preview

3

Approve

Gate for production

4

Deploy

The stack with GitHub Actions or Azure Pipelines

5

Authenticate

Workload identity federation, no stored secrets

Azure Policy guardrails

- Require encryption settings, secure transfer, and minimum TLS
- Deny public network access or require private endpoints
- Require immutability or backup coverage for designated data classes
- Require tags, or inherit them, for cost allocation
- Group definitions into initiatives and track compliance reporting

Policy effects to know

Effect	What it does	Storage example
Audit	Reports non-compliance	Accounts without private endpoints
Deny	Blocks non-compliant requests	Public blob access enabled
Modify	Adds or changes properties	Add a missing tag
DeployIfNotExists	Deploys a related resource	Enable backup for a resource



Demonstration: the whole platform in minutes

1. Deploy a research storage landing zone from one Bicep module set: account, lifecycle, immutability, Backup vault, RBAC, tags
2. Show tags flowing into a Cost Management view by research unit
3. Show an Azure Policy assignment flagging a deliberately non-compliant account

A governed, costed platform delivered automation first, with a view a finance partner would recognize.



Break - 15 minutes

Back for the capstone roadmap workshop.

BLOCK 3

Capstone roadmap workshop and presentations



The capstone: a two-year hybrid data platform roadmap

01

Scope

Working as one team, cover storage, archival, protection, DR, database modernization, and FinOps

02

Sequence

Order initiatives across four half-year phases

03

Fund

Identify the first three initiatives to fund, the cost model behind each, and the decisions leadership must make

From nearly all on-premises to hybrid

Quick wins

Low-risk, high-value moves such as archive and backup targets

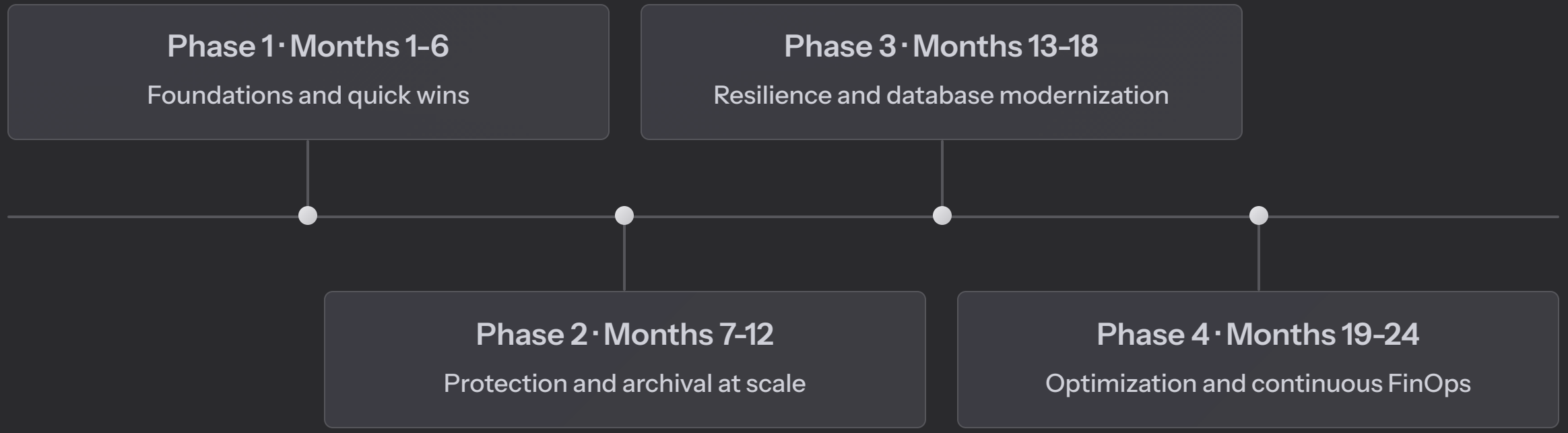
Dependencies

Identity, connectivity, governance, and tagging first

Decision points

Where leadership approval, budget, or policy is required

Roadmap phases at a glance



A starting frame; teams adjust content per phase.

Candidate initiatives drawn from the course



Tagged, lifecycle-managed research storage with reserved capacity



Century-proof archival landing zone with integrity validation



Immutable backup target for on-premises protection



University resilience matrix applied to six workload classes



Immutability tier for the Oracle estate



Bicep and Azure Policy platform with Cost Management showback

How the 105 minutes run

01

Frame

Agree scope and phases on the roadmap template

02

Place

Initiatives and dependencies

03

Cost

Select and cost the first three initiatives

04

Decide

List leadership decisions

05

Present

Share the roadmap and receive instructor feedback

What makes a roadmap leadership will fund

- Each initiative tied to a risk reduced or a cost saved
- Honest cost lines and stated assumptions
- Clear owners and decision dates
- Options, not a single answer



BLOCK 4

Course wrap-up and survey



Three days in review

01

Architect storage at petabyte scale with a cost rationale

02

Archive for the long term with immutability and integrity

03

Protect data in both directions against ransomware and insiders

04

Match resilience and DR to each workload class

05

Protect Oracle and enterprise databases with immutability

06

Govern it all as code with FinOps built in

Keep going: materials and next steps

Materials

Demo script, Bicep module set and pipeline definition, storage cost model workbook, roadmap template

Certification pathway

Certification pathway reference: **AZ-104** Azure Administrator Associate (prerequisite for the Expert credential), **AZ-305** Azure Solutions Architect Expert, **DP-300** Azure Database Administrator Associate. For security-focused staff, **SC-500** replaced **AZ-500** in August 2026.

Microsoft Learn

Content was aligned to Microsoft Learn; re-verify features and limits as the platform evolves.

Course survey

Please complete the course survey.